

# HE2003 Econometrics I

## Quiz Practice 1 (Solutions)

AY 2026/2027, Semester 1

*Quantitative Research Society @NTU*

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### Overview

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- **Coverage:** Lectures 1–5 and Lecture 6 slides 1–3 (large-sample distribution of  $\hat{\beta}_1$  and its standard error); no t-tests, confidence intervals, p-values or Stata.
- **Format:** 5 questions, 50 marks total, approximately 60 minutes.
- **Standard normal table** ( $\Phi(z) = P(Z \leq z)$  for  $Z \sim N(0, 1)$ ), needed only where a probability is asked:

| $z$       | 0.50   | 1.00   | 1.28   | 1.50   | 1.645  | 1.96   | 2.00   | 2.50   | 2.58   | 3.00   |
|-----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $\Phi(z)$ | 0.6915 | 0.8413 | 0.8997 | 0.9332 | 0.9500 | 0.9750 | 0.9772 | 0.9938 | 0.9951 | 0.9987 |

Use  $P(Z > z) = 1 - \Phi(z)$  and, by symmetry,  $\Phi(-z) = 1 - \Phi(z)$ .

- **Question mix:** Q1 derivation (OLS FOCs and residual identities); Q2 regression-output interpretation (prediction, change in prediction, units); Q3 large-sample sampling distribution of  $\hat{\beta}_1$  (standardising, probability calculations, the role of  $SE(\hat{\beta}_1)$ ); Q4 binary regressor (group means, an OLS-with-a-dummy proof, Least Squares Assumption 1 in this setting); Q5 conceptual (causality, Least Squares Assumption 1, selection bias, Type I/II trade-off).

## Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

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**Question 1****(10 Marks)**

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n.$$

For candidate values  $b_0, b_1$ , define the sum of squared errors

$$S(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2.$$

The OLS estimators  $\hat{\beta}_0, \hat{\beta}_1$  are the values of  $b_0, b_1$  that minimize  $S(b_0, b_1)$ .

- (a) (4 marks) Derive the two first-order conditions for this minimization problem, and show that they can be written as

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.$$

- (b) (3 marks) Using part (a), show that the OLS residuals  $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$  satisfy

$$\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n \hat{u}_i X_i = 0.$$

- (c) (3 marks) Using part (b), show that  $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$ , where  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$  is the OLS predicted value.

**Solution****Method 1: Direct approach**

- (a) *First-order conditions.* Differentiate  $S(b_0, b_1)$  with respect to  $b_0$  and  $b_1$  and set the derivatives to zero at  $(b_0, b_1) = (\hat{\beta}_0, \hat{\beta}_1)$ :

$$\left. \frac{\partial S}{\partial b_0} \right|_{(\hat{\beta}_0, \hat{\beta}_1)} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0,$$

$$\left. \frac{\partial S}{\partial b_1} \right|_{(\hat{\beta}_0, \hat{\beta}_1)} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.$$

Dividing both equations by  $-2$  gives the stated first-order conditions:

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.$$

- (b) *Residual identities.* By definition, the OLS residual is  $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ . Substituting this definition directly into the two first-order conditions from part (a) gives

$$\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n \hat{u}_i X_i = 0.$$

(The first says the residuals sum to zero; the second says the residuals are uncorrelated with  $X$  in the sample.)

(c)  $\sum \hat{u}_i \hat{Y}_i = 0$ . Expand  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$  inside the sum:

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i) = \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i.$$

By part (b), both  $\sum_{i=1}^n \hat{u}_i = 0$  and  $\sum_{i=1}^n \hat{u}_i X_i = 0$ , so

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \hat{\beta}_0(0) + \hat{\beta}_1(0) = \boxed{0}.$$

### Marking scheme (10 marks)

(a) **(4 marks)** First-order conditions

- 2 marks: correct partial derivative of  $S$  with respect to  $b_0$ , set to zero.
- 2 marks: correct partial derivative of  $S$  with respect to  $b_1$ , set to zero, and both simplified (divided by  $-2$ ) to the stated form.

(b) **(3 marks)** Residual identities

- 1 mark: correctly identifies  $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$  as the object appearing in the FOCs.
- 2 marks: correctly substitutes to obtain both  $\sum_{i=1}^n \hat{u}_i = 0$  and  $\sum_{i=1}^n \hat{u}_i X_i = 0$  (1 mark each).

(c) **(3 marks)**  $\sum \hat{u}_i \hat{Y}_i = 0$

- 1 mark: correctly expands  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$  and splits the sum.
- 1 mark: correctly invokes both identities from part (b).
- 1 mark: correct final conclusion  $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$ .

## Question 2

(10 Marks)

A researcher studying the Singapore private rental market collects data on  $n = 150$  HDB flat rental listings and regresses monthly rent (in SGD) on floor area (in square metres, sqm). The estimated OLS regression line is

$$\widehat{\text{Rent}}_i = 380 + 24.5 \text{ Area}_i.$$

- (3 marks) A 90 sqm flat is listed for rent. What is the regression's predicted monthly rent for this flat?
- (3 marks) A tenant moves from a 70 sqm flat to a 95 sqm flat. What is the regression's predicted change in monthly rent?
- (4 marks) Interpret the slope coefficient, including units. Is the intercept economically meaningful in this context? Explain briefly.

## Solution

### Method 1: Direct approach

- Predicted rent at Area = 90.* Substitute Area = 90 into the regression line:

$$\widehat{\text{Rent}} = 380 + 24.5(90) = 380 + 2205 = 2585.$$

$\widehat{\text{Rent}} = \text{SGD } 2,585 \text{ per month.}$

- Predicted change in rent.* Since the regression line is linear in Area, the predicted change equals the slope times the change in Area,  $\Delta \text{Area} = 95 - 70 = 25$ :

$$\Delta \widehat{\text{Rent}} = \hat{\beta}_1 \times \Delta \text{Area} = 24.5 \times 25 = 612.5.$$

(Check:  $\widehat{\text{Rent}}(70) = 380 + 24.5(70) = 2095$ ;  $\widehat{\text{Rent}}(95) = 380 + 24.5(95) = 2707.5$ ; difference = 612.5.)

$\Delta \widehat{\text{Rent}} = \text{SGD } 612.50 \text{ (an increase).}$

- Interpretation.* The slope  $\hat{\beta}_1 = 24.5$  means: each additional square metre of floor area is associated with an average increase of **SGD 24.50 per month** in predicted rent. The intercept  $\hat{\beta}_0 = 380$  is the model's prediction at Area = 0 sqm; since no rental flat has zero floor area, this value has **no direct economic meaning** – it is simply the value needed to fit the line to the data over the observed range of Area, exactly as with STR = 0 in the Test Score example from lecture.

Slope: +SGD 24.50/month per extra sqm; intercept not economically meaningful.

### Marking scheme (10 marks)

- (3 marks)** Predicted rent
  - 1 mark: correct substitution Area = 90 into the regression line.
  - 1 mark: correct arithmetic.
  - 1 mark: final answer stated with correct units (SGD/month).

(b) **(3 marks)** Predicted change

- 1 mark: correct method (slope  $\times$  change in Area, or difference of two predictions).
- 1 mark: correct arithmetic ( $\Delta\text{Area} = 25$ , product = 612.5).
- 1 mark: correctly stated with units and direction (increase).

(c) **(4 marks)** Interpretation

- 2 marks: correct slope interpretation with units and correct sign/direction.
- 2 marks: correct discussion that the intercept is not economically meaningful (Area=0 does not occur in the data), with brief justification.

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### Question 3

(10 Marks)

A team of researchers studies  $n = 250$  Singapore tech start-ups, regressing monthly revenue growth (%) on monthly R&D spending (SGD '000s). Based on the large sample, they treat the OLS estimator  $\hat{\beta}_1$  as following its large-sample sampling distribution

$$\hat{\beta}_1 \sim N(\beta_1, \sigma_{\hat{\beta}_1}^2),$$

with population slope  $\beta_1 = 2.0$  and standard deviation of the sampling distribution  $\sigma_{\hat{\beta}_1} = \text{SE}(\hat{\beta}_1) = 0.5$ .

- (2 marks) Explain, in words, why  $\hat{\beta}_1$  – computed from this one sample of 250 start-ups – is itself a random variable, even though the population slope  $\beta_1$  is a fixed (if unknown) number.
- (3 marks) Using the large-sample result  $\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1)$ , standardise and use the standard normal table in the Overview to compute  $P(\hat{\beta}_1 > 2.75)$ .
- (3 marks) Compute  $P(|\hat{\beta}_1 - \beta_1| < 0.98)$ , i.e. the probability that the estimator falls within 0.98 of the true population slope. Use the standard normal table in the Overview.
- (2 marks) Explain what  $\text{SE}(\hat{\beta}_1)$  measures, and give two reasons – referring to the formula for  $\sigma_{\hat{\beta}_1}^2$  – why it becomes *smaller* as (i) the sample size  $n$  increases, and (ii) the sample spread of the regressor  $X$  increases.

### Solution

#### Method 1: Direct approach

- $\hat{\beta}_1$  as a random variable.  $\hat{\beta}_1$  is computed from the particular sample of 250 start-ups that happened to be drawn. A different random sample of 250 start-ups from the same population would contain different firms and hence different  $(Y_i, X_i)$  values, which would plug into the OLS formula  $\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$  and produce a different numerical estimate. Because its value depends on which random sample was drawn,  $\hat{\beta}_1$  varies from sample to sample and so has a probability distribution (its *sampling distribution*) even though the population parameter  $\beta_1$  it estimates is a fixed constant.

$\hat{\beta}_1$  is a function of the random sample  $(Y_i, X_i)_{i=1}^n$ , so it varies across samples and is itself random.

- $P(\hat{\beta}_1 > 2.75)$ . Standardise using  $\beta_1 = 2.0$  and  $\sigma_{\hat{\beta}_1} = 0.5$ :

$$P(\hat{\beta}_1 > 2.75) = P\left(\frac{\hat{\beta}_1 - 2.0}{0.5} > \frac{2.75 - 2.0}{0.5}\right) = P(Z > 1.5),$$

where  $Z \sim N(0, 1)$  in large samples. From the table,  $\Phi(1.5) = 0.9332$ , so

$$P(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668.$$

$$P(\hat{\beta}_1 > 2.75) = 0.0668.$$

(c)  $P(|\hat{\beta}_1 - \beta_1| < 0.98)$ . Divide through by  $\sigma_{\hat{\beta}_1} = 0.5$ :

$$P(|\hat{\beta}_1 - 2.0| < 0.98) = P\left(\left|\frac{\hat{\beta}_1 - 2.0}{0.5}\right| < \frac{0.98}{0.5}\right) = P(|Z| < 1.96).$$

By symmetry of the standard normal distribution,

$$P(|Z| < 1.96) = 2\Phi(1.96) - 1 = 2(0.9750) - 1 = 0.9500.$$

$$P(|\hat{\beta}_1 - \beta_1| < 0.98) = 0.9500.$$

(d) *What  $\text{SE}(\hat{\beta}_1)$  measures, and why it shrinks.*  $\text{SE}(\hat{\beta}_1)$  is an estimate of  $\sigma_{\hat{\beta}_1}$ , the standard deviation of the sampling distribution of  $\hat{\beta}_1$ : it measures how much  $\hat{\beta}_1$  would typically vary from one random sample to another (the spread/dispersion of the sampling distribution), i.e. how precisely  $\hat{\beta}_1$  pins down the true  $\beta_1$ . Recalling

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \cdot \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 u_i^2}{\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]^2},$$

- **(i) Larger  $n$ :** the explicit factor  $1/n$  in front, and the fact that a bigger sample contains more independent pieces of information about the relationship between  $X$  and  $Y$ , makes  $\sigma_{\hat{\beta}_1}^2$  shrink as  $n$  grows – more data means a more precisely estimated slope.
- **(ii) More spread in  $X$ :**  $\sigma_{\hat{\beta}_1}^2$  has  $[\sum_{i=1}^n (X_i - \bar{X})^2]^2$ -type terms in the denominator, i.e. it is decreasing in the sample variance of  $X$ . When the regressor  $X_i$  takes a wider range of values, the data trace out the regression line over a longer horizontal stretch, which pins down the slope more precisely (compare: if all  $X_i$  were nearly identical, almost any line through the cloud of points would fit about equally well).

$\text{SE}(\hat{\beta}_1)$  measures the spread of  $\hat{\beta}_1$ 's sampling distribution;  
it falls as  $n$  rises and as the spread of  $X$  rises.

### Marking scheme (10 marks)

(a) **(2 marks)** Why  $\hat{\beta}_1$  is random

- 1 mark: recognizes  $\hat{\beta}_1$  is a function of the random sample  $(Y_i, X_i)$ .
- 1 mark: correctly explains that a different sample gives a different numerical value, hence a sampling distribution, while  $\beta_1$  stays fixed.

(b) **(3 marks)**  $P(\hat{\beta}_1 > 2.75)$

- 1 mark: correct standardisation,  $z = 1.5$ .
- 1 mark: correctly reads  $\Phi(1.5) = 0.9332$  from the table.
- 1 mark: correct final probability 0.0668.

(c) **(3 marks)**  $P(|\hat{\beta}_1 - \beta_1| < 0.98)$

- 1 mark: correct standardisation to  $P(|Z| < 1.96)$ .

- 1 mark: correct use of symmetry,  $2\Phi(1.96) - 1$ .
- 1 mark: correct final probability 0.9500.

(d) **(2 marks)** Meaning of  $SE(\hat{\beta}_1)$  and why it shrinks

- 1 mark: correctly states  $SE(\hat{\beta}_1)$  measures the spread/dispersion of the sampling distribution of  $\hat{\beta}_1$  (precision of the estimator).
- 1 mark: valid explanation of *both* why it falls with larger  $n$  and with more spread in  $X$ .

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## Question 4

(10 Marks)

A study of  $n = 200$  food-delivery riders in Singapore regresses monthly income (in SGD) on  $D_i$ , a dummy variable equal to 1 if the rider owns their vehicle and 0 if the rider rents it ( $n_0 = 120$  renters,  $n_1 = 80$  owners). The estimated regression is

$$\widehat{\text{Income}}_i = 1450 + 320 D_i, \quad \text{SE}(\hat{\beta}_1) = 95.$$

- (3 marks) State the average monthly income predicted by the regression for renters and for owners.
- (2 marks) Give an economic interpretation of  $\hat{\beta}_1 = 320$ .
- (3 marks) Prove, using the OLS normal equations from Question 1 applied to this binary regressor  $D_i \in \{0, 1\}$ , that in general  $\hat{\beta}_0 = \bar{Y}_0$  (the sample mean of income among renters) and  $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$  (the difference in sample mean income between owners and renters), where  $\bar{Y}_0, \bar{Y}_1$  are the group sample means and  $n_0, n_1$  are the group sizes.
- (2 marks) State what Least Squares Assumption 1 ( $E[u_i | D_i] = 0$ ) requires in this dummy-regressor setting, give one specific unobserved factor that could plausibly make it fail here, and state the likely direction of the resulting bias in  $\hat{\beta}_1$ .

## Solution

### Method 1: Direct approach

- Group means.* Since  $D_i \in \{0, 1\}$ ,  $E[\text{Income}_i | D_i] = \hat{\beta}_0 + \hat{\beta}_1 D_i$  (sample analogue). For renters ( $D = 0$ ):

$$\widehat{\text{Income}}|_{D=0} = 1450 + 320(0) = 1450.$$

For owners ( $D = 1$ ):

$$\widehat{\text{Income}}|_{D=1} = 1450 + 320(1) = 1770.$$

Renters: SGD 1,450/month; Owners: SGD 1,770/month.

- Interpretation of  $\hat{\beta}_1$ .* Because  $D$  is binary,  $\hat{\beta}_1$  is exactly the difference in average monthly income between the two groups:

$$\hat{\beta}_1 = \widehat{\text{Income}}|_{D=1} - \widehat{\text{Income}}|_{D=0} = 1770 - 1450 = 320.$$

Riders who own their vehicle earn, on average, SGD 320/month more than those who rent.

- Proof that  $\hat{\beta}_0 = \bar{Y}_0$  and  $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$ .* From Question 1, the OLS estimators satisfy the two normal equations

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 D_i) = 0, \quad \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 D_i) D_i = 0.$$

**First equation.** Expanding and using  $\sum_{i=1}^n D_i = n_1$  (the number of owners,  $D_i = 1$ ):

$$\sum_{i=1}^n Y_i = n\hat{\beta}_0 + n_1\hat{\beta}_1, \tag{i}$$

where  $n = n_0 + n_1$ . Splitting  $\sum_{i=1}^n Y_i$  by group,  $\sum_{i=1}^n Y_i = n_0 \bar{Y}_0 + n_1 \bar{Y}_1$ .

**Second equation.** Since  $D_i \in \{0, 1\}$ ,  $D_i^2 = D_i$ , so  $\sum_{i=1}^n D_i^2 = n_1$ . Also  $\sum_{i=1}^n Y_i D_i$  sums  $Y_i$  only over the owner group, i.e.  $\sum_{i=1}^n Y_i D_i = n_1 \bar{Y}_1$ . The second normal equation becomes

$$n_1 \bar{Y}_1 - \hat{\beta}_0 n_1 - \hat{\beta}_1 n_1 = 0 \implies \bar{Y}_1 = \hat{\beta}_0 + \hat{\beta}_1. \quad (\text{ii})$$

**Combining.** Substitute  $n_1 \bar{Y}_1 = n_1 \hat{\beta}_0 + n_1 \hat{\beta}_1$  (from (ii)) into  $n_0 \bar{Y}_0 + n_1 \bar{Y}_1 = n \hat{\beta}_0 + n_1 \hat{\beta}_1$  (from (i)):

$$n_0 \bar{Y}_0 + n_1 \hat{\beta}_0 + n_1 \hat{\beta}_1 = n \hat{\beta}_0 + n_1 \hat{\beta}_1 \implies n_0 \bar{Y}_0 = (n - n_1) \hat{\beta}_0 = n_0 \hat{\beta}_0.$$

Since  $n_0 > 0$ , dividing gives  $\hat{\beta}_0 = \bar{Y}_0$ ; substituting back into (ii) gives  $\hat{\beta}_1 = \bar{Y}_1 - \hat{\beta}_0 = \bar{Y}_1 - \bar{Y}_0$ .

$\hat{\beta}_0 = \bar{Y}_0 \text{ (renter mean),} \quad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0 \text{ (owner mean - renter mean).}$

- (d) *Least Squares Assumption 1 in this setting.* Here Assumption 1,  $E[u_i | D_i] = 0$ , requires that the average of the unobserved determinants of income is zero *within each group*:  $E[u_i | D_i = 0] = 0$  for renters *and*  $E[u_i | D_i = 1] = 0$  for owners, i.e. owners and renters must not differ, on average, in any unobserved factor that also affects income.

A plausible violation: riders who *choose* to own their vehicle may on average be more experienced or more committed to the job (e.g. they have worked longer and know the most profitable delivery routes/times) – an unobserved factor correlated with ownership that would raise their income even without the ownership effect itself. This self-selection means  $E[u_i | D_i = 1] > E[u_i | D_i = 0]$ , so Assumption 1 fails.

**Direction of bias:** since the omitted factor (experience/commitment) is positively correlated with both ownership and income,  $\hat{\beta}_1$  is **biased upward** – part of the estimated SGD 320 gap reflects pre-existing differences between the two groups rather than a pure effect of ownership.

$E[u | D] \neq 0 \text{ (self-selection into ownership by experience/commitment)}$   
 $\implies \hat{\beta}_1 \text{ is upward biased.}$

### Marking scheme (10 marks)

- (a) **(3 marks)** Group means

- 1 mark: correct method ( $D = 0$  and  $D = 1$  substituted into the regression line).
- 1 mark: correct renter mean (1450).
- 1 mark: correct owner mean (1770).

- (b) **(2 marks)** Interpretation of  $\hat{\beta}_1$

- 1 mark: recognizes  $\hat{\beta}_1$  as the difference in group means.
- 1 mark: correct direction/magnitude with units (owners earn SGD 320 more).

- (c) **(3 marks)** Proof that  $\hat{\beta}_0 = \bar{Y}_0$ ,  $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$

- 1 mark: correctly uses the first normal equation (splitting  $\sum_{i=1}^n Y_i$  by group and using  $\sum_{i=1}^n D_i = n_1$ ).

- 1 mark: correctly uses the second normal equation, including  $D_i^2 = D_i$ , to obtain  $\bar{Y}_1 = \hat{\beta}_0 + \hat{\beta}_1$ .
- 1 mark: correct algebra combining the two to reach  $\hat{\beta}_0 = \bar{Y}_0$  and  $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$ .

(d) **(2 marks)** Least Squares Assumption 1

- 1 mark: correctly states Assumption 1 as requiring  $E[u_i \mid D_i = 0] = 0$  and  $E[u_i \mid D_i = 1] = 0$  separately.
- 1 mark: valid confounding factor with correct direction of bias.

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## Question 5

(10 Marks)

- (a) (4 marks) A researcher regresses PSLE (Primary School Leaving Examination) score on weekly hours of private tuition, using observational data on Singapore primary-six pupils. She finds a strong positive coefficient on tuition hours and concludes that private tuition *causes* higher PSLE scores. Using Least Squares Assumption 1 ( $E[u_i|X_i] = 0$ ) and the concept of selection bias, explain why this conclusion may not be valid. In your answer, state a specific unobserved factor that could be correlated with tuition hours, and predict the likely direction of the resulting bias.
- (b) (3 marks) Suggest one research design that would let the researcher obtain a more credible estimate of the causal effect of tuition on PSLE score, and briefly explain why it addresses the problem identified in part (a).
- (c) (3 marks) A bank's automated loan-approval system flags each application as "high risk" (reject) or "safe" (approve) based on a credit-score cutoff. Define, in the context of this bank's decision, what a Type I error and a Type II error would mean, and explain the trade-off the bank faces if it raises the credit-score cutoff required for approval.

## Solution

### Method 1: Direct approach

- (a) *Selection bias in the tuition/PSLE regression.* Write the model as

$$\text{PSLE}_i = \beta_0 + \beta_1 \text{Tuition}_i + u_i.$$

Least Squares Assumption 1 requires  $E[u_i | \text{Tuition}_i] = 0$  for *every* level of tuition hours – i.e., pupils who receive more tuition should, on average, have the same unobserved factors as pupils who receive less.

In reality, an unobserved factor such as **parental income and educational motivation** (or the child's own innate academic ability) is a plausible component of  $u_i$  that is **correlated with tuition hours**: wealthier and more academically engaged families are both more likely to purchase many hours of private tuition *and* more likely to have children who would score well regardless (better home academic support, higher ability). This is a textbook case of **selection bias** (self-selection into tuition based on factors that also directly affect PSLE score), so  $E[u_i | \text{Tuition}_i]$  is increasing in tuition hours and Assumption 1 is violated.

**Direction of bias:** since the omitted factor is positively correlated with both tuition hours and PSLE score,  $\hat{\beta}_1$  will be **biased upward** – part of the "effect" the researcher attributes to tuition is really the effect of family resources/motivation/ability that would have raised the score anyway.

|                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|
| $E[u   \text{Tuition}] \neq 0 \text{ (selection on family resources and ability)}$ $\Rightarrow \hat{\beta}_1 \text{ is upward biased.}$ |
|------------------------------------------------------------------------------------------------------------------------------------------|

- (b) *A more credible research design.* One option: a **randomized controlled trial (RCT)** in which free tuition slots are randomly assigned to a subset of eligible pupils (or a natural experiment/instrumental-variable design exploiting quasi-random variation in access to tuition, e.g. a subsidy voucher allocated by lottery).

**Why this works:** random assignment makes  $\text{Tuition}_i$  statistically independent of all pre-existing unobserved characteristics (family income, motivation, ability) that make up  $u_i$ . Full independence implies  $E[u_i \mid \text{Tuition}_i] = E[u_i] = 0$ , so Assumption 1 is restored by construction and the tuition/no-tuition comparison of average PSLE scores identifies the causal effect  $\beta_1$ .

Random assignment of tuition  $\Rightarrow E[u \mid \text{Tuition}] = 0$   
 $\Rightarrow$  causal interpretation valid.

(c) *Type I/Type II errors and the cutoff trade-off.* In this context:

- **Type I error:** the bank flags a truly **safe** applicant (one who would have repaid) as “high risk” and rejects them – a lost lending/interest-income opportunity.
- **Type II error:** the bank flags a truly **high-risk** applicant as “safe” and approves the loan, which later defaults – a direct financial loss.

**Trade-off:** raising the credit-score cutoff makes approval harder. This screens out more genuinely risky applicants, **lowering the Type II error rate** (fewer defaults slip through), but it also rejects more applicants who would actually have repaid, **raising the Type I error rate**. Because both error types depend on the same single cutoff, the bank cannot reduce both simultaneously by moving the threshold alone – it must trade one off against the other (or invest in a better screening model to shift the whole trade-off).

Higher cutoff:  $\downarrow$  Type II error (fewer bad loans),  
 $\uparrow$  Type I error (more good applicants rejected).

### Marking scheme (10 marks)

(a) **(4 marks)** Selection bias

- 1 mark: correctly states Assumption 1 ( $E[u \mid X] = 0$ ) as the relevant condition.
- 1 mark: identifies a specific, plausible confounding factor (e.g. family income, motivation, or ability).
- 1 mark: explains why the confounder is correlated with tuition hours (self-selection).
- 1 mark: correct direction of bias (upward) with valid justification.

(b) **(3 marks)** Alternative design

- 1 mark: proposes a valid design (RCT, natural experiment, or instrumental variable).
- 1 mark: explains that random/quasi-random assignment breaks the correlation between tuition and  $u$ .
- 1 mark: correctly links this back to Assumption 1 being restored / causal interpretation being valid.

(c) **(3 marks)** Type I/II trade-off

- 1 mark: correct Type I error definition in context.
- 1 mark: correct Type II error definition in context.
- 1 mark: correct description of the trade-off when the cutoff is raised.